Overview of image security techniques with applications in multimedia systems

Raymond B. Wolfgang and Edward J. Delp

Video and Image Processing Laboratory (*VIPER*)
School of Electrical and Computer Engineering
Purdue University
West Lafayette, Indiana, 47907-1285
USA

ABSTRACT

The growth of networked multimedia systems has created a need for the copyright protection of digital images and video. Copyright protection involves the authentication of image content and/or ownership. This can be used to identify illegal copies of a (possibly forged) image. One approach is to mark an image by adding an invisible structure known as a digital watermark to the image. Techniques of incorporating such a watermark into digital images include spatial-domain techniques, transform-domain algorithms and sub-band filtering approaches.

1 INTRODUCTION

The recent growth of networked multimedia systems has increased the need for the protection of digital media. This is particularly important for the protection and enforcement of intellectual property rights. Digital media includes text, digital audio, images, video and software. Many approaches are available for protecting digital data; these include encryption, authentication and time stamping. In this paper we present algorithms for image authentication and forgery prevention known as watermarks. Figure 1 shows the block diagram for watermarking digital images.

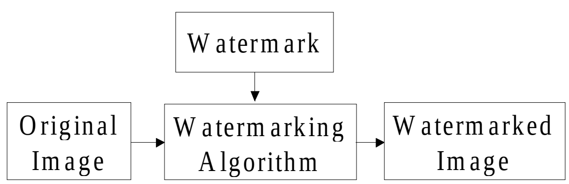

Figure 1. Block diagram of a watermarking algorithm

1.1 Basic Scenario for Image Protection

A user has created an electronic image at some effort and expense, and wants to make it available on a communications network. When unauthorized copies or forgeries of the image appear elsewhere on the network, the user needs to prove that the image belongs to them. One also needs to determine if and by how much the image has been changed from the original. Image protection techniques must provide:

1. Copy detection to identify unauthorized copies of an image.
2. Content authentication to verify the content of a copy of an image, since the copy may have been forged or filtered.

The ability to time stamp an image is important, as is a method to determine an image's chain of custody. This verifies who viewed (or altered) an image, when it was viewed and in what order. It is also desirable to have some ability to determine where in an image changes have been made.

This work was partially supported by a grant from the AT&T foundation. Address all correspondence to E.J. Delp, ace@ecn.purdue.edu, http://www.ece.purdue.edu/~ace, or +1 765 494 1740.

1.2 Video

Authenticated video is needed for many scenarios. Video-on-demand (VOD) systems must prevent unauthorized viewing along with illegal copying and distribution. For digital video rentals, a user may download the video from a server. To implement an electronic rental period, the digital video sequence must incapacitate itself after the period expires. This requires a reliable third-party time source to provide time stamping. A new possibility is to rent digital video for a fixed number of viewings over a longer time period. Table 1.1 describes several video applications, with the cryptographic tools each would require. Cryptography for digital video is further discussed in [1].

Table 1.1. Scenario list for video security

Scenario	Encrypt.	Auth.	T.S.
Closed Circuit TV surveillance of a money access machine.		X	X
Generate an audit or viewing history of a video		X	X
Video-on-Demand / Video Server Applications:	X	X	X
Prevent playing the video past the rental expiration			X
Limit the number of times the movie can be played	X	X	
Prevent playing the video unless provider is legitimate		X	

1.3 Cryptographic Tools

Many tools are available to protect images from unauthorized viewing, copying, distribution and modification. These include encryption, authentication and time stamps [2,3]. Encryption disguises the content of digital media. Only users who possess the decryption "key" can convert the encrypted data back to its original form. Without the key, it is computationally infeasible to derive the original data. Encryption and authentication are not the same. Authentication does not hide the content of the data, but proves who created the document. Time stamps pinpoint the data's owner *and* the time at which the data was generated. One trivial way to prove ownership of an image is to own the earliest reliable time stamp of the image. All three tools may be used in various combinations.

1.3.1 Encryption

Encryption is a one-to-one mapping of a sequence of symbols from a given set to another sequence of symbols, usually from the same set [2]. The original data, P, is known as the *plaintext*; the encrypted data is the *ciphertext*, C. C is a function of P and the enciphering *key*, K_{ENC}.

$$C = F(P, K_{ENC}) \tag{1}$$

$$P = F(C, K_{DEC}) \tag{2}$$

To obtain P from C, one needs the deciphering key, K_{DEC}. It is extremely difficult to obtain P from C unless one has K_{DEC}. In *symmetric* encryption, $K_{ENC} = K_{DEC}$. The Data Encryption Standard (DES) [3] is an example of a symmetric algorithm used often in industry. Public key cryptography (e.g. the *RSA algorithm*) is an example of *asymmetric* encryption; one key is publicly distributed ($K_{ENC} = K_{PUB}$), while the other is kept private ($K_{DEC} = K_{PRI}$). The sender would encrypt P with the recipient's K_{PUB}. Only the recipient possesses K_{PRI} , which is needed to decrypt C.

$$C = F(P, K_{PUB}) \tag{3}$$

$$P = F(C, K_{PRI}) \tag{4}$$

Encryption will not prevent legitimate purchasers from altering an image after decryption.

1.3.2 Authentication

Public key cryptography can authenticate data without encrypting them. To authenticate a data set, one encrypts it with one's own K_{PRI} . Anyone can then decrypt C with the owner's K_{PUB}. A *digital signature* is a form of authentication. Digital signatures and time stamps make use of *hash functions* [2]. A hash function has as its input an arbitrary length binary P, and obtains a fixed length binary output H.

$$H = H(P) \tag{5}$$

H is a function of *all* the input bits. Changing a single input bit will change the output significantly. There are several useful properties of hash functions:

1. It is computationally trivial to reproduce *H*, given *P*.
2. It is extremely difficult to reproduce *P*, given only *H*.
3. It is very difficult to find two inputs P_1 and P_2 that produce the same *H*.

Many different hash functions exist; MD-4 and MD-5 are very popular, as is the US National Institute of Standards and Technology (NIST) Secure Hash Standard, SHA-1. SHA-1 is used in the NIST Digital Signature Standard. These and additional algorithms are detailed in [2,3].

1.3.3 Time Stamps

One way to authenticate an image is to prove that one was the first to possess it. For this, one needs some way to "stamp" the image with the time of creation much like the Post Office stamps mail with a postmark. Forward or back dating of the time stamp must not be possible. Let X be the data to be time stamped, and Y = H(X) be the hash of X. An owner sends an official request, R, to a third party time stamping service (TSS).

$$R_n = (Y_n, I_n) \tag{6}$$

The TSS then produces a certificate, C_n:

$$C_n = (n, t_n, I_n, Y_n; L_n) \tag{7}$$

$$L_n = (t_{n-1}, I_{n-1}, Y_{n-1}, H(L_{n-1})) \tag{8}$$

The request number is n; t_n is the time of the request and I_n is the owner's identification string. L_n is known as the linking string and is a concatenation of the previous request time, identification string, document hash and linking string hash. The TSS waits for the next request, I_{n+1}, then concatenates it to C_n. The time stamp returned to the user is

$$S = F((C_n, I_{n+1}), K_{PRI}) \tag{9}$$

where K_{PRI} is the private key of the TSS and F is the encryption function. A challenger to the time stamp first authenticates the TSS signature on the stamp. The challenger can then ask I_{n-1} (embedded in L_n) to show her time stamp. The quantities in L_n must match those in C_{n-1}. The hash of her L_{n-1} must also match the value of $H(L_{n-1})$ present in L_n. If any value does not match, one of the two time stamps is false. In practice, the linking string contains information from the last N requests. This distributes the authentication responsibility, since any one of the N requesters may verify the time stamp. This process may continue backwards or forwards to the challenger's satisfaction. Improvements to this procedure are discussed in [2].

These are the tools and basic concepts used in digital watermarking.

2 LITERATURE REVIEW

An image may be authenticated with what is known as a digital watermark. A watermark is a secret code or image incorporated into an original image which acts to verify both the owner and content of the image. The use of perceptually invisible watermarks is one form of image authentication. A watermarking algorithm consists of three parts:

1. Watermark
2. Marking algorithm
3. Verification algorithm

Each owner has a unique watermark. The marking algorithm incorporates the watermark into the image. The verification algorithm authenticates the image, determining both the owner and the integrity of the image.

One early watermarking method obtains a checksum of the image data, then embeds the checksum into the LSB of randomly chosen pixels[4]. Others add a modified maximal length linear shift register sequence to the pixel data. They identify the watermark by using the spatial crosscorrelation function of the modified sequence and part of the watermarked image [5,6,7]. Watermarks also can modify the image's spectral or transform coefficients directly. These algorithms often modulate DCT coefficients according to a sequence known only to the owner [8,9]. Watermarking techniques may be image dependent. These techniques increase the level of the watermark in the image while maintaining the imperceptibility of the mark [10,11,12]. One method incorporates features from most of the above techniques [13,14]. Its wavelet implementation lends itself to watermarking data rate-scalable video [15]. Time stamps thwart a clever attack proposed by IBM [16] on all of these watermarking schemes. Visible watermarks also exist; IBM has developed a proprietary visible watermark to protect images that are part of the digital Vatican library project [17]. The watermarking itself is only a small part of any controlled access and distribution scheme; a method for secure distribution would combine encryption with digital watermarking [18]. The sections below describe these watermarking algorithms in detail; many of these techniques may be used in combination with each other.

2.1 Checksum Technique

This watermark is formed from the checksum value of the seven most significant bits of all pixels [4]. A checksum is the modulo-2 addition of a sequence of fixed-length binary words; it is a special type of hash function. In this technique, one word is the concatenation of eight 7-bit segments, which come from eight different pixels. Each pixel is involved in the checksum only once. The final checksum is fifty-six bits. The technique then randomly chooses the locations of the pixels that are to contain one bit of the checksum. The pixel locations of the checksum, together with the checksum itself, form the watermark, *W*. The last bit of each chosen pixel is changed (if necessary) to equal the corresponding checksum bit. *W* must be kept secret. To verify this watermark the checksum of a test image *Z* is obtained, and compared to the ideal version in W. Any discrepancy invalidates *Z*. The advantages and disadvantages of this technique are as follows.

Advantages

1. Embedding *W* only changes (on average) half of the pixels covered by *W*. This not only reduces visual distortion but also increases security.
2. An image may hold many *W* as long as they do not overlap.
3. This method is very fast.

Disadvantages

1. This watermarking method is fragile. Any change to either the image data itself or the embedded checksum can cause the verification procedure to fail.
2. The checksum method does not detect pixel swaps or similar attacks. A forger could replace a section of Y_7 with one of equal size and checksum.
3. An attacker could remove the entire watermark by replacing the LSB plane.

2.2 Basic M-Sequence Approach

This watermark is based on using a modified m-sequence [5]. A linear feedback shift register with n stages can form pseudo-random binary sequences with periods as large as $2^n - 1$. M-sequences achieve this maximum period, and have a very desirable autocorrelation and randomness properties [19]. Two types of sequences may be formed from an m-sequence: *unipolar* and *bipolar*. The elements of a bipolar sequence are {-1,1} and the elements of a unipolar sequence are {0,1}. In [5], *X* is a grayscale 512 x 512 image, and *w* is an extended one-dimensional bipolar m-sequence of length 512. *W* consists of 512 randomly shifted copies of *w* - one for each row in *X*. *W* is then arithmetically added to *X* to form the watermarked image, *Y*.

$$Y = X + W \tag{10}$$

To verify a possibly forged image row *z* relative to the original row *y*, the spatial crosscorrelation function is obtained.

$$R_{zw}(\alpha) = \sum_j [z(j) - E[z]]\, w(j-\alpha) \tag{11}$$

E[z] is the average pixel value of row *z*. The presence of the peak in R_{zw} is determined. If there is no peak, *z* is not authentic.

Advantages

1. This watermark is robust to small amounts of noise introduced in the image. (The peak of R_{zw} is still large.)
2. Pixel swaps require knowledge of *W*.
3. Multiple watermarks can overlap; successive watermarks treat the previously watermarked image as a new *X*.

Disadvantages

1. The last two bit planes could be removed and replaced.
2. An attacker can deduce *W* if *2n* consecutive bits in *W* are known.
3. This method does not specifically protect the DC value of the pixels covered by an individual block.

2.3 Two-Dimensional Spatial Watermarks

If one particular row is forged, the above method can accurately identify which row has been changed. It is preferable, however, to localize errors in two dimensions. This can easily be achieved with using a two-dimensional watermark blocks, and testing an image on a block-by-block basis. An example is the Variable-W Two-Dimensional Watermark (VW2D) [6,7].

The watermark in VW2D is created as follows:

1. A bipolar m-sequence with a period of $2^{96} - 1$ is obtained, and the first 128 bits are discarded.
2. The next 64 bits are shaped column-wise into an 8 x 8 block, *w*. The next 32 bits are discarded. This step repeats to form additional *w*.

$$y = x + w \tag{12}$$

y is the watermarked image block. This process is repeated until the entire image *X* is marked. The total number of watermark blocks is image dependent; together they form the watermark *W*. To verify a possibly forged image block *z*, one must obtain the crosscorrelation function:

$$R_{zw}(\alpha,\beta) = \sum_x \sum_y z(x,y)\, w(x-\alpha, y-\beta) \tag{13}$$

$$\delta = R_{yw}(0,0) - R_{zw}(0,0) \tag{14}$$

If $\delta < T$, where *T* is the test threshold, *z* is genuine. Large values of *T* tolerate changes to the marked image block *y*. If $z = y$, then $\delta = 0$. The advantages of this technique over the basic m-sequence approach are as follows.

1. The level of tolerance to changes is user-determined by adjusting *T*.
2. The verification algorithm is much less computationally intensive.
3. Changes in an image can be localized to a certain block of the image.

For 24-bit color images, each color plane may be treated as a monochrome image. Another possibility is to add *W* to the first color plane, then add encrypted versions of *W* to the subsequent planes. For YUV images, one may choose to only add *W* to the *Y* plane (luminance). Three-dimensional watermarks are also possible. For VW2D, a simple Yes/No threshold may be expanded to several levels [7]: unaltered, slightly affected, definitely altered but still belonging to the owner, and completely changed or not belonging to the tested owner. One disadvantage remains with both techniques. If an attacker knows the watermark bit, $w(i,j)$, for two pixels, those two pixels can be changed by equal amounts in opposite directions without changing δ. The attacker can successfully change other pixels for which $w(i,j)$ is unknown as long as δ remains below *T*. One may not have to know the entire *w* to successfully change a reasonably-sized portion of *y*. Figure 2 and Figure 3 show an original and watermarked version of the Tia image. Figure 4 shows the VW2D watermark.

2.4 DCT Coefficient Modulation

An algorithm that places a watermark in the most perceptually significant areas of an image is described in [8]. The marking procedure modulates the discrete cosine transform (DCT) coefficients of the image using a one-dimensional watermark, *W*. The technique is robust to many types of image distortions (including cropping, very low bit-rate JPEG compression and D/A conversion), as well as collusion from several independently watermarked images.

W is a sequence of normally distributed, zero-mean unit-variance random numbers. A DCT transform is performed on the entire image, and the transform coefficients are then modulated as follows. Let *X* be the original image, *Y* be the watermarked image, and X_D and Y_D be the row-concatenated DCT coefficients of *X* and *Y* respectively. Let $X_D(i)$ and $Y_D(i)$ be the i^{th} DCT coefficient in X_D and Y_D. $W(i)$ is the i^{th} element in the watermark sequence; *a* is a scale factor which prevents unreasonable values for $Y_D(i)$. The marking is then performed:

$$Y_D(i) = X_D(i)(1 + aW) \tag{15}$$

Figures 5-7 show the Tia image marked with a = 0.1, 0.5 and 1.0 respectively. Other algorithms to incorporate *W* into X_D are suggested in [8]. Inversely transforming Y_D to form *Y* completes the marking procedure. The first step of the verification procedure is to obtain a copy of *W* from a possibly forged watermarked image, *Z*. Z_D is the row-concatenated vector of *Z*'s DCT coefficients. Let W^* be the extracted version of *W*.

$$W^*(i) = \frac{1}{a}\left[\frac{Z_D(i)}{X_D(i)} - 1\right] \tag{16}$$

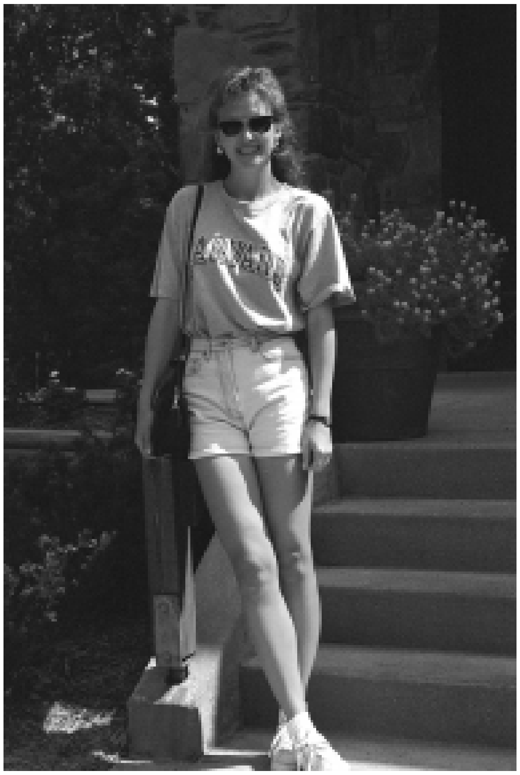

Figure 2. Original Tia Image

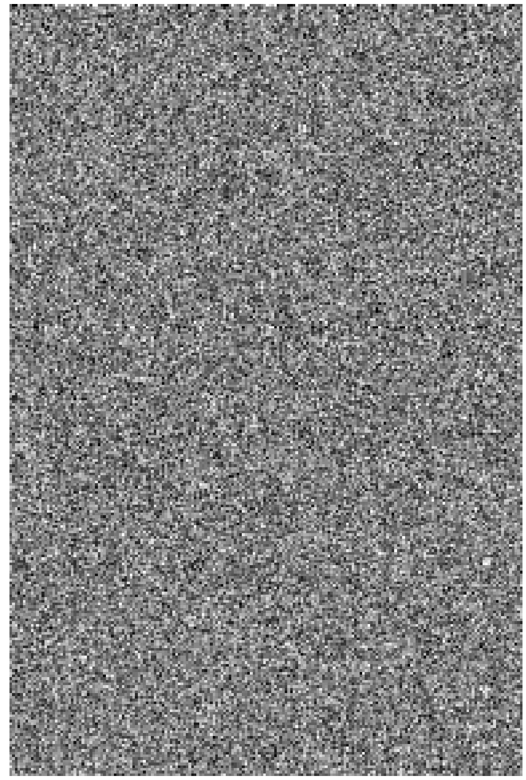

Figure 4. Watermark for VW2D

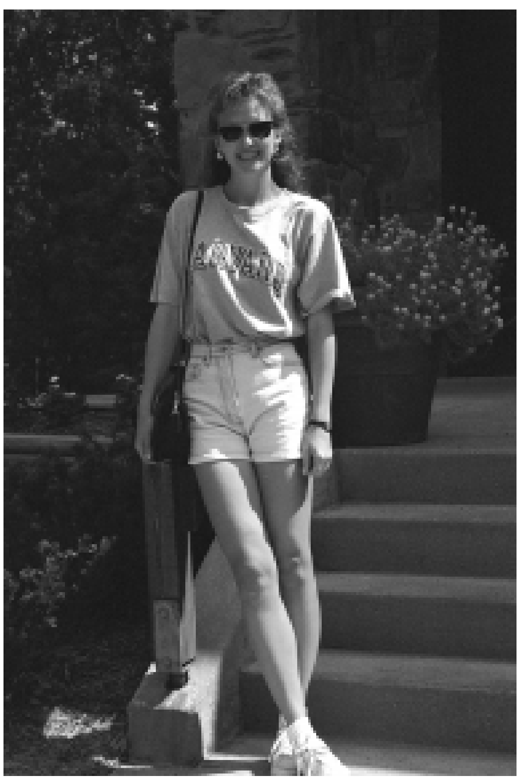

Figure 3. Watermarked Tia Image

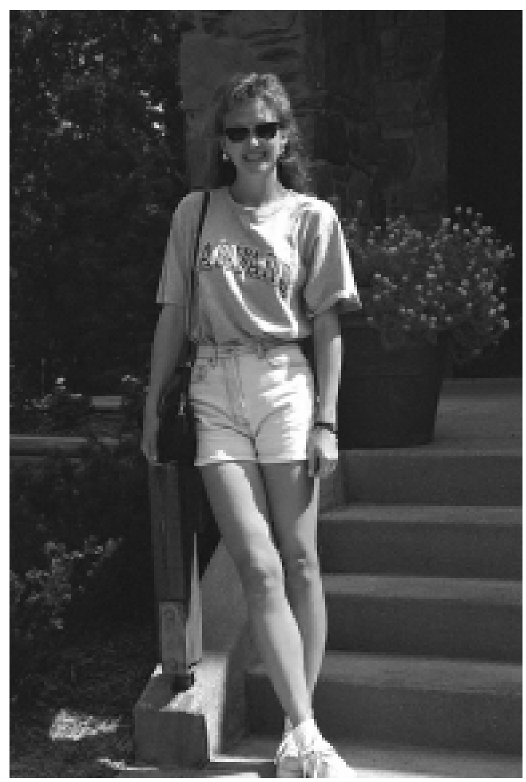

Figure 5. Tia image watermarked with a = 0.1

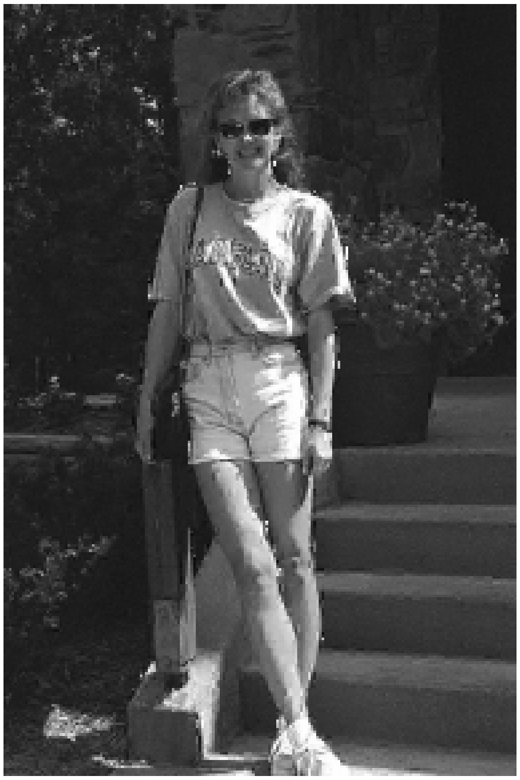

Figure 6. Tia image watermarked with a = 0.5

Figure 7. Tia image watermarked with a = 1.0

A measure of similarity between W^* and the original W is defined as:

$$S(W,W^*) = \frac{W \cdot W^*}{\sqrt{W^* \cdot W^*}} \tag{17}$$

If an image has not been watermarked with W, S is a zero mean random variable. If W^* differs only slightly from W, then $E[S] >> 0$. A hypothesis test on S can determine if W is indeed the image's watermark. This technique accommodates multiple watermarks, and a much wider range of attacks than the m-sequence techniques. The effect of embedding W, however, is often visible, as Figures 6-7 show. This can be corrected with masking techniques [13].

A method very similar to [8] also modulates DCT coefficients, but uses a one-dimensional bipolar binary sequence for W [10]. The DCT of the original image is first obtained. The marking procedure is:

1. The DCT coefficients are sorted according to their absolute magnitude.
2. A percentage of total energy, P, is defined by the owner.
3. The largest n coefficients that make up P percent of the total energy in the image are identified. The watermark will be added only to the non-DC coefficients in this list.

For all i such that $X_D(i)$ is one of the selected coefficients,

$$Y_D(i) = X_D(i) + W(i) \tag{18}$$

In this way the watermark is image dependent. A larger P increases the number bits of W that can be embedded in X, but it increases the chance that W will be perceptible. W and the list of selected coefficients must be kept secret. The verification procedure first extracts W^* from Z_D by subtracting Z_D from X_D. The remaining steps are the same as in the previous technique. The image is marked on a block-by-block basis. Note that [8] and [10] require both W and X to extract the watermark, whereas the sub-band and spatial techniques only require W. The spatial techniques, however, only verify the watermark's presence; nothing is formally extracted.

2.5 Sub-Band Watermarking Technique

This algorithm is described in [11]. An original image is first passed through a Gabor filter bank; other filter banks may be used [20]. The energy in each band is obtained as shown in Figure 8. The center frequencies of each filter are indicated as f_k. E_k is the energy of the image present in the frequency band specified by filter k. The next step is to obtain a random two-dimensional array which is then low-pass filtered to form G. G is then modulated at each center frequency, f_k to form G_k. The energy of G_k , D_k, is obtained. The watermark can then be formed:

$$W = \sum_{k=1}^{n} \alpha_k W_k \tag{19}$$

where $\alpha_k = 1$ if $E_k \geq D_k$, and $\alpha_k = 0$ if $E_k < D_k$. Then

$$Y = X + W \tag{20}$$

The verification procedure first passes Y through the filter bank. Then compute the crosscorrelation function between each Y_k and G_k for k such that $\alpha_k = 1$. From this crosscorrelation one can determine if G_k is actually present in Y_k. If a sufficient number of peaks are present at a sufficient strength, the image belongs to the owner of W. An advantage of this watermark is that it occupies a significant part of the image, rather than just the lower order bits. This technique is compatible with JPEG compression, and tolerates common watermark attacks such as the addition of noise and low-pass filtering.

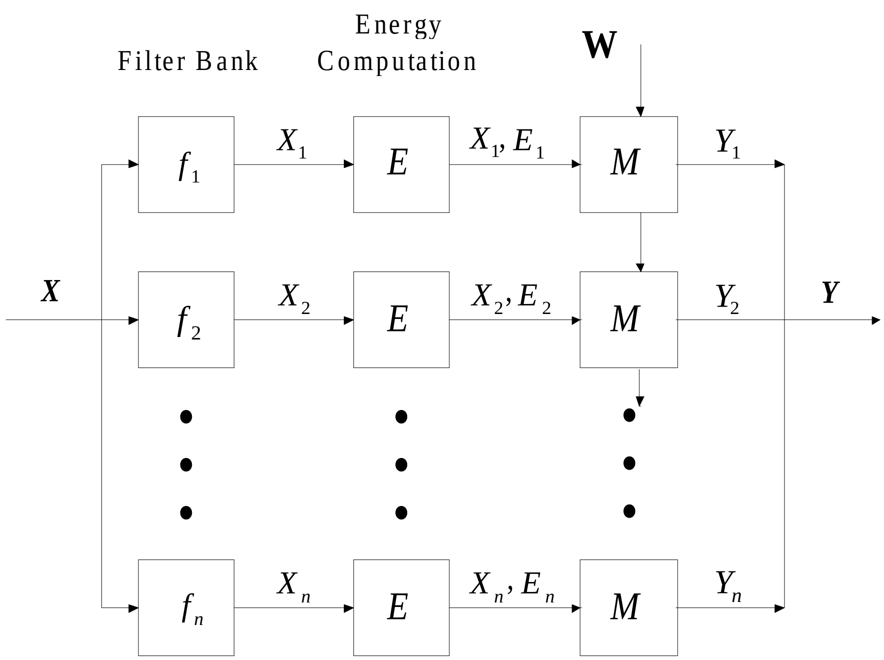

Figure 8. Block diagram of sub-band watermarking algorithm.

2.6 Visual Models

An algorithm that has many of the advantages of the above techniques is described in [13,14]. Crosscorrelation functions are used in the verification procedure; the maximum of these functions are compared to ideal maximums, much like in VW2D. The embedding procedure uses visual models (masking) to determine if the watermark will be visible. These visual models, which result in image-*dependent* DCT or Wavelet quantization matrices, were first developed for image compression [21]. The result is a watermark that is amplified in regions where such a change would not be noticeable, and set to zero in regions where any addition to the image would be perceived, as in the sub-band technique. This method is also robust against cropping, re-scaling and scanning/reprinting.

The watermark is a one-dimensional sequence of N(0,1) random numbers; this sequence is inserted directly into the DCT coefficients of an 8 x 8 image block x_D of an image as follows:

$$y_D(i) = \begin{cases} x_D(i) + w(i)\ jnd(i), & x_D(i) > jnd(i) \\ x_D(i), & else \end{cases} \tag{21}$$

$y_D(i)$ are the marked transform coefficients, and the marked image block y is obtained from the inverse DCT transform. $jnd(i)$ is the *just-noticeable difference* value that corresponds to the ith coefficient in x_D. The *jnd* values are based on the particular image's quantization matrix. If x_D is larger than its *jnd* value, then x_D can tolerate the addition of a watermark component. Larger *jnd* values signify x_D's larger tolerance to variations (and therefore watermark bits with greater amplitude). In this way the watermark strength adapts to the local image content. The explanation of the *jnd* values requires a brief discussion of visual masking and its connection to quantization noise in an image.

Contrast masking is the phenomenon whereby one signal makes another similar signal at lower amplitude imperceptible. If two notes of similar frequency are played on two different violins, but one is played at a significantly lower volume, the softer one will not be heard at all. The louder note "masks" the softer tone. The same phenomenon occurs when viewing two similar (in spatial frequency) contrast gratings of different intensity. For the one grating to be seen, its contrast must be significantly higher than the other. Less similar gratings are visible together at much larger difference in contrast [22].

This masking characteristic of the human eye helps images mask quantization noise. In fact, an image can be non-uniformly quantized to a fairly large signal-to-noise ratio (SNR), but still appear acceptable. It is this exact technique that JPEG compression exploits with its quantization matrix. It identifies the quantization step allowable for each individual DCT coefficient; this can be interpreted as the amount of error a given DCT coefficient can have without producing unacceptable visual distortion. This also determines the maximum amplitude of w that will not perceptually distort x. Image-dependent matrices can identify regions where the addition of the watermark to the particular image would be visible, and where the amplitude of w could be greater without being noticed. Adjusting w's strength makes w more robust: harder to remove, yet less visible.

The verification algorithm extracts the watermark, then uses a threshold test on a single test statistic to determine the entire image's authenticity. First, W^* , which is a row-concatenated version of each block-based watermark w^*, is extracted from the test image's DCT coeffients, Z_D:

$$W^* = X_D - Z_D \tag{22}$$

Each watermark bit is then scaled by the corresponding *jnd* value, and a normalized correlation coefficient is then formed. A traditional hypothesis test is performed on ρ. This quantity will be zero-mean, normally distributed if W and W^* are independent. In vector form:

$$W^* = \frac{W^*}{JND} \tag{23}$$

$$\rho_{ww^*} = \frac{W \cdot W^*}{W \cdot W} \tag{24}$$

$$\begin{aligned} &\rho_{ww^*} > T \quad Z\ authentic \\ &\rho_{ww^*} \leq T \quad Z\ not\ authentic \end{aligned} \tag{25}$$

T is a user-defined threshold based on the desired probability of watermark detection and probability of false alarm. The DCT implementation lends itself to directly embedding the watermark into the JPEG bit stream. This makes watermarking huge image libraries much more efficient [14]. The wavelet version is suited for marking scalable video [15]. Another DCT-based technique also employs masking, but marks the original image with W_D, the DCT of W.

Tewfik [12] describes a method that performs the DCT on W before embedding it in the original image. The amplitude of W can then be easily adjusted to ensure its invisibility. W is now a two-dimensional reshaped m-sequence, formed from a multitude of 8 x 8 watermark blocks w. The image is marked on a block-by-block basis. The two-dimensional DCT of w, w_D is multiplied by a scaling matrix m before being added to x_D:

$$y_D = x_D + m \cdot w_D \tag{26}$$

Figure 9 shows the block diagram for this method. y is examined to ensure that the watermark is imperceptible; if not, m is attenuated and the original image block is re-watermarked. M is the entire array of masking blocks m.

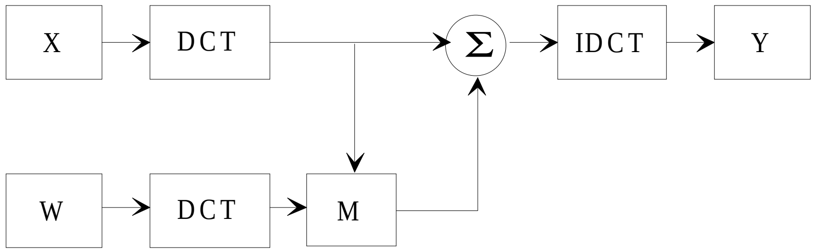

Figure 9. Block diagram of spectral method with scaling.

The watermark verification procedure is similar to the spatial methods. A spatial crosscorrelation function is obtained between the entire *W* and the zero-mean version of the entire test image *Z*. A hypothesis test is then performed on the maximum of this crosscorrelation. If the result is above a certain threshold, *T*, the image belongs to the owner of the watermark. Testing may be performed on sub-areas of *Z* (as in VW2D) to localize changes to the image.

2.7 A Visible Watermark

IBM has developed a proprietary visible digital watermark for artwork in the Vatican library [17]. The Vatican required a watermarking technique that detects any forgeries made to an image, but does not alter the image content to a point where the research value of the image is compromised. The first step in this technique is to choose a watermark, *W*. This may be a picture of a known trademark; it does not have to be a noise-like image, nor the same size as *X*. Pixels in *W* are specified to be either “active” or “transparent”. An image sub-block, *x*, that is the same size as W is extracted. If the watermark pixel is classified as transparent, the corresponding pixel in the image is not adjusted. For active watermark pixels, the image pixel’s color information is changed by a small but visible amount. The algorithm for producing the change is kept secret. This process is repeated until the entire image is marked. The watermark appears as a distinct visible pattern present throughout the image; the luminance plane of the image remains unaltered, which preserves the image’s research value. Verification is performed by inspection; forgeries made to the image will erase the color adjustment and produce a visible break in the pattern. In this way the visible watermark offers forgery and copy detection, but does not degrade the content of the image. This technique is compatible with JPEG compression.

3 THE IBM ATTACK

An attack on a wide class of watermarking schemes is proposed in [16]:

$$Y_1 = X + W_1 \tag{27}$$

$$X_F = Y_1 - W_2 \tag{28}$$

$$\Rightarrow Y_1 = X_F + W_2 \tag{29}$$

X_F is known as the “counterfeit original”. The watermarked image, Y_1, now appears to be a marked version of X_F, marked with W_2. It cannot be discerned which is the actual original, *X* or X_F. Note that the counterfeit original was created *without* access to the true original, *X*.

One solution is to time stamp *X* when it is created. Then, the owner with the earliest-dated original is the true owner. A watermark that is a function of a non-invertible (one-way) function of *X* would also securely determine ownership. Let *H* be the hash of *X* [2].

$$H = H(X) \tag{30}$$

A time stamp, *S*, would then be used as the seed for generating *W*. *T* is the time of *X*’s creation, and *U* is the owner name.

$$S = (H, T, U) \tag{31}$$

$$Y_1 = X + W(S) \tag{32}$$

It is computationally infeasible to obtain the correct watermark without knowing *S*. To successfully steal Y_1, one must find an X_F such that

$$X_F = Y_1 - W(S_F) \tag{33}$$

Since $W(S_F)$ depends on X_F, this task is unrealistic.

4 ACCESS CONTROL ON THE WORLD WIDE WEB

The marking and verification of an image is only one tool to address the problem of copyright infringement. A system which also regulates image distribution would greatly reduce the number of illegal copies on a network. This is particularly important in the context of the Internet, where the copying of images is easily performed with current browser software. A security system which works within popular browsers would be even more useful. An image security system that works as a plug-in to Netscape Navigator is proposed in [18]. Customers must first register with the company that owns the desired multimedia; this includes providing a public key (K_{PUB}) to authenticate future requests and purchases. This database is accessible to the data server. A user would first request an image (for instance) from the server with the specialized Navigator plug-in. The plug-in encrypts both the request and a time stamp of the request with the user's K_{PRI}, and sends this message to the server. The server then decrypts the message with the user's K_{PUB}; if the request is deemed authentic, a modified copy of the requested image is sent to the client.

The unmarked original cannot be sent to the customer; this would allow unrestrained distribution of the original work. The original image X is first watermarked with the user's own W. The marked image Y is partially *encrypted* to form Y_E; Y_E contains all the data in Y, but is commercially useless. The image is symmetrically encrypted with the TIE (Tools for Image Encryption) suite [18]. These methods result in blurred images, images with excerpts removed, or other effects which damage, but do not destroy an image. Only upon payment is the decryption key *for the image* securely sent to the user. With this key alone the user can obtain Y. The entire operation is performed through the plug-in, which prevents intermediate printing or saving of the Y without payment. Figure 10 shows the block diagram for the procedure of obtaining Y_E.

5 CONCLUSION

We have overviewed many of the recent techniques in the digital watermarking of images. Other important topics include the protection of documents, audio and video. While all of the techniques are not foolproof, it is important that they help preserve the “chain of evidence of ownership” that is consistent with intellectual property law.

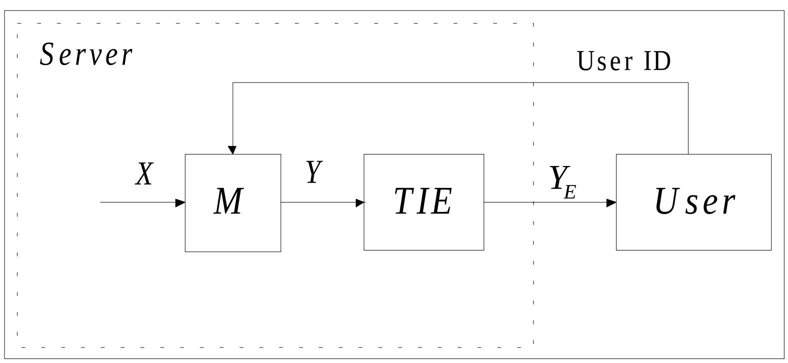

Figure 10 Block diagram of a server releasing Y_E instead of Y.

REFERENCES

1. Benoit M. Macq and Jean-Jacques Quisquater, “Cryptology for digital TV broadcasting,” *Proceedings of the IEEE*, vol. 83, no. 6, June 1995, pp. 944-957.
2. Bruce Schneier, *Applied Cryptography*, Second Ed., Wiley & Sons, 1996.
3. Douglas R. Stinson, *Cryptography, Theory and Practice*, CRC Press, 1995.
4. Steve Walton, “Information authentication for a slippery new age,” *Dr. Dobbs Journal*, vol. 20, no. 4, April, 1995, pp. 18-26.
5. R. G. van Schyndel, A.Z.Tirkel, N.R.A Mee, C.F. Osborne, “A digital watermark,” *Proceedings of the IEEE International Conference on Image Processing*, November, 1994, Austin, Texas, USA, vol. 2, pp. 86-90.

6. Raymond B. Wolfgang and Edward J. Delp, “A watermark for digital images,” *Proceedings of the 1996 International Conference on Image Processing*, Lausanne, Switzerland, Sept. 16-19, 1996, vol. 3, pp. 219-222.
7. Raymond B. Wolfgang and Edward J. Delp, “Techniques for watermarking digital imagery: further studies”, *Proceedings of the International Conference on Imaging Science, Systems, and Technology*, Las Vegas, Nevada, USA, June 30 – July 3, 1997, vol. 1, pp. 279-287.
8. Ingemar J. Cox, Joe Kilian, Tom Leighton and Talal Shamoon, “Secure spread spectrum watermarking for images, audio and video,” *Proceedings of the 1996 International Conference on Image Processing*, Lausanne, Switzerland, Sept. 16-19, 1996, vol. 3, pp. 243-246.
9. E. Koch and J. Zhao, “Towards robust and hidden image copyright labelling,” *Proceedings of the 1995 IEEE Workshop on Nonlinear Signal and Image Processing*, Neos Marmaras, Greece, June 20-22, 1995, pp. 452-455.
10. F. M. Boland, J. J. K. Ó Ruanaidh and C. Dautzenberg, “Watermarking digital images for copyright protection,” *Proceedings of the International Conference on Image Processing and its Applications*, Edinburgh, Scotland, July 1995, pp. 321-326.
11. J.-F. Delaigle, C. De Vleeschouwer, B. Macq, “Digital watermarking of images,” *Proceedings of the IS&T/SPIE Symposium on Electronic Imaging Science and Technology*, 1996.
12. Mitchell D. Swanson, Bin Zhu and Ahmed H. Tewfik, “Transparent robust image watermarking,” *Proceedings of the 1996 International Conference on Image Processing*, Lausanne, Switzerland, Sept. 16-19, 1996, vol. 3, pp. 211-214.
13. Christine I. Podilchuk and Wenjun Zeng, “Digital image watermarking using visual models”, *Proceedings of the IS&T/SPIE Conference on Human Vision and Electronic Imaging II*, San Jose, CA, USA, Feb. 10-13, 1997, vol. 3016, pp. 100-111.
14. Christine I. Podilchuk and Wenjun Zeng, “Watermarking of the JPEG bitstream,” *Proceedings of the International Conference on Imaging Science, Systems, and Technology*, Las Vegas, Nevada, USA, June 30 – July 3, 1997, pp. 253-260.
15. K. Shen and E. J. Delp, “A Control Scheme for a Data Rate Scalable Video Codec,” *Proceedings of the IEEE International Conference on Image Processing*, Lausanne, Switzerland, Sept. 16-19, 1996, vol. 2, pp. 69-72.
16. Scott Craver, Nasir Memon, Boon-Lock Yeo and Minerva Yeung. “Can invisible watermarks resolve rightful ownerships?,” *Proceedings of the IS&T/SPIE Conference on Storage and Retrieval for Image and Video Databases V*, San Jose, CA, USA, Feb. 13-14, 1997, vol. 3022, pp. 310-321.
17. Fred Mintzer, Albert Cazes, Francis Giordano, Jack Lee, Karen Magerlein and Fabio Schiattarella, “Capturing and preparing images of Vatican library manuscripts for access via Internet,” *Proceedings of the IS&T 48th Annual Conference*, Washington, DC, USA, May, 1995, pp. 74 – 77.
18. D. Storch and E. Koch, “Controlable user access on multimedia data in the world wide web,” *Proceedings of the International Conference on Imaging Science, Systems, and Technology*, Las Vegas, Nevada, USA, June 30 – July 3, 1997, pp. 270-278.
19. Henry Beker and Fred Piper, *Cipher Systems: The Protection of Communications*, Northwood Publications, 1982.
20. P.P. Vaidyanathan, *Multirate Systems and Filter Banks*, Prentice-Hall, 1993.
21. A. B. Watson, “DCT quantization matrices visually optimized for individual images,” *Proceedings of the SPIE Conference on Human Vision, Visual Processing and Digital Display IV*, 1992, pp. 202-216.
22. A. B. Watson, “The cortex transform: rapid computation of simulated neural images,” *Computer Vision, Graphics, and Image Processing*, 1987, pp. 311-327.